

Measuring what fMRI adds

ADHD-200 / Python · PyTorch · scikit-learn · nilearn

PROBLEM

A multi-site fMRI benchmark can learn site, motion or demographic patterns.
I separated representation quality from incremental information beyond controls.

METHOD

Audited 378 participants; 5 repeats x 3 subject-level folds; 201/51/126 roles.
Compared Pearson and tangent connectivity with LR and a 64/16 MLP.
Training-only feature selection, scaling and tangent reference; validation-only hyperparameters, early stopping and three-seed averaging.

RESULTS

Pearson MLP 0.6096 -> tangent MLP 0.6453: +0.0357 AUC, positive in 5/5 repeats.
Full non-imaging control 0.7077 -> control + image predictions 0.7127.
The +0.0050 incremental gain was small and partition-sensitive (4/5 positive).

ENGINEERING EVIDENCE

Independent replay of 35,910 frozen test rows and 60 validation fusion weights.
Preserved scientific source hashes; portable CLI and explicit environment checks.
Tests for subject/role leakage, label alignment and training-only preprocessing.
Runnable synthetic LR/MLP demo with checkpoint replay, curves and timing.

HIRING RELEVANCE

Medical AI: imaging QC, covariate controls and limits on site generalization.
DS / MLE: data contracts, reproducibility, numerical debugging and model tradeoffs.

SCOPE AND OWNERSHIP

Mentor-led project with AI-assisted development/review. Internal reused cohort;
no diagnostic or externally validated prediction claim. Repeat ranges are not CIs.
This release replays evidence and prepares real splits; full training and raw-image reconstruction were not rerun. Historical 409-person/structural studies are separate.